AIVA: 40% Milestone Technical Report

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Abstract:

This project presents an enterprise microservice deployment leveraging Docker containers and Kuber

FastAPI is employed as the high-throughput asynchronous backend web framework, communicating v

ChromaDB serves as the localized vector database, paired with Google Gemini text embeddings for d

For exact technical term retrieval, Okapi BM25 algorithm is utilized to catch sparse keywords.

Reciprocal Rank Fusion with a rank constant k=60 balances semantic similarity against lexical keywor

Redis is integrated as an in-memory caching tier to achieve sub-50ms latency.

The primary bottleneck observed during stress testing was database connection pool exhaustion unde